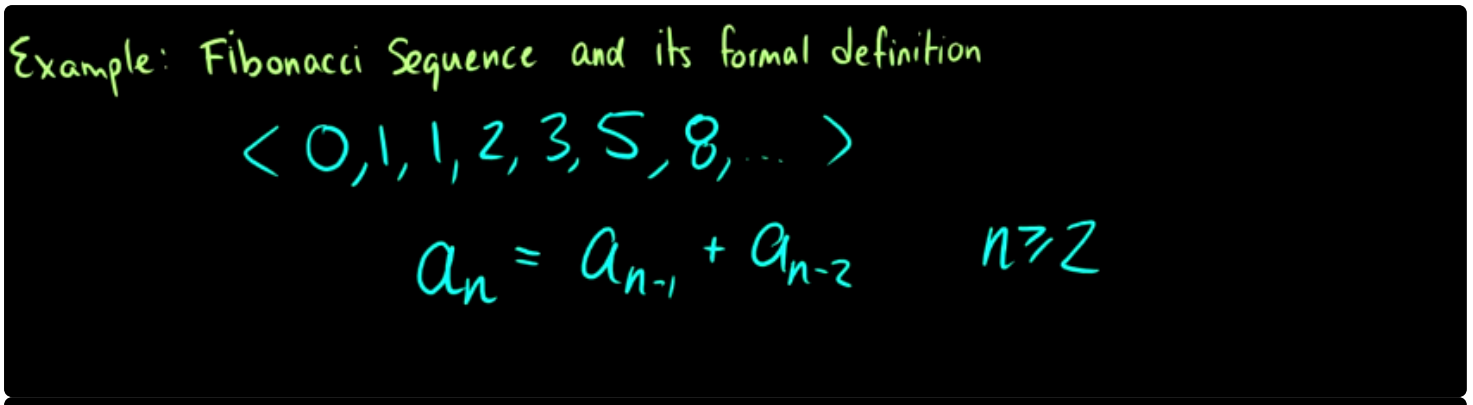
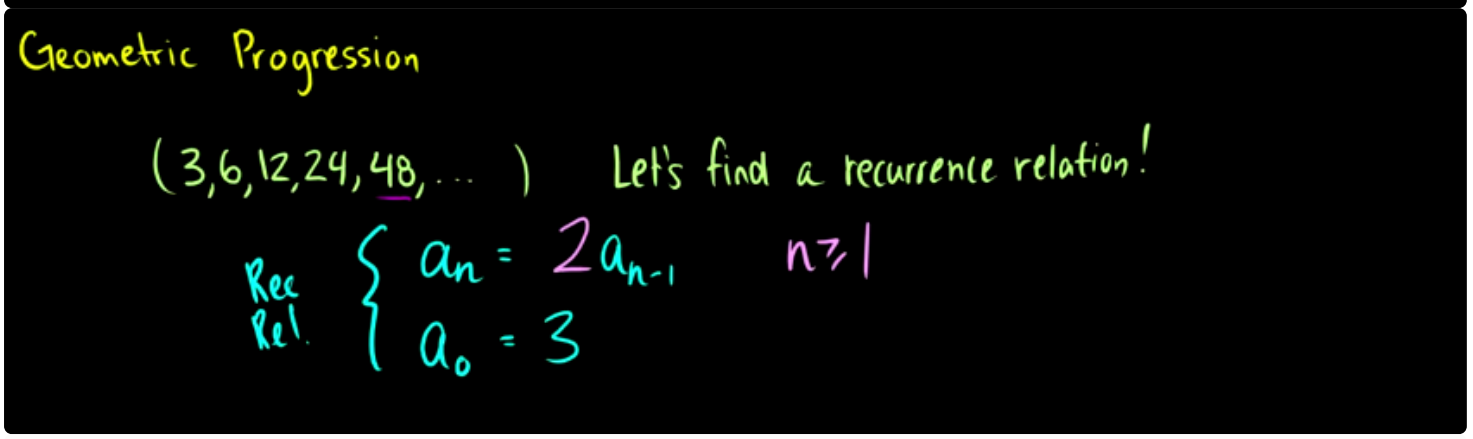
unit-ii-recurrence-relations
Introduction

A recurrence equation relates the value, a_n, of a sequence in terms of some or all of its past values, a_{n-1}, a_{n-2}, ...
Solving a recurrence equation means to find a close-form of the function defined by the recurrence equation.





Example 8.1 $a_1 = 2$, and $a_n = 4a_{n-1} - 2$ for $n \geq 2$.

Example 8.2 $a_1 = 2, a_2 = 7$, and $a_n = a_{n-1}^2 + 2a_{n-2}$ for $n \geq 3$.

Example 8.3 $a_1 = 1, a_2 = 2, a_3 = 1$, and $a_n = 3a_{n-1} - a_{n-2} + 2a_{n-3}$ for $n \geq 4$.

Linear recurrence relation

A recurrence equation for a_n is called linear if it can be written as a linear function of its past values, a_{n-1} , a_{n-2} , etc.

Example 8.1 above is a linear recurrence equations, whereas Example 8.2 is a nonlinear recurrence equation because a_n is a nonlinear function of a_{n-1}.

A recurrence is **linear** if a_n is expressed as a **linear combination** of the previous terms — meaning each previous term $a_{n-1}, a_{n-2}, \ldots$ appears only to the **first power**, multiplied by a coefficient, and added together. No previous term is squared, multiplied by another previous term, put inside another function, etc.

General Form

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} + f(n)$$

where:

- $c_1, c_2, \ldots, c_k$ are coefficients (constants, or possibly functions of n — see note below)

Order of recurrence relation

The order of a linear recurrence equation is k if an is a linear function of k past values an−1 , an−2 , . . . , an−k . In some cases, all of the past values may not be present on the left-hand side. For this reason the order of the recurrence equation is defined as the difference between the largest and smallest subscripts of the equation.

Part 1: The Basic Idea of Order

The **order** of a linear recurrence is how far back into the sequence's history the formula reaches to compute a_n.

If a_n depends on $a_{n-1}, a_{n-2}, \ldots, a_{n-k}$, the order is k.

Example (order 1):

$$a_n = 3a_{n-1}$$

Only reaches back one step → order 1.

Example (order 2):

$$a_n = a_{n-1} + a_{n-2} \quad \text{(Fibonacci)}$$

Reaches back two steps → order 2.

Example (order 3):

$$a_n = 4a_{n-1} - 2a_{n-2} + a_{n-3}$$

Reaches back three steps → order 3.

So far this matches intuition: order = "how many previous terms appear."

Part 2: The Subtlety — When a Term in the Middle Is Missing

Here's where the second half of the definition matters. Sometimes the recurrence **skips** a term — it uses a_{n-1} and a_{n-3}, say, but *not* a_{n-2}. If you just counted "how many terms appear," you might undercount. The correct rule is:

$$\text{order} = (\text{largest subscript gap}) - (\text{smallest subscript gap})$$

...measured from a_n itself. Equivalently, look at the subscript **indices** used (as a distance from n) and take the difference between the largest and smallest.

Example — a missing middle term:

$$a_n = 2a_{n-1} + 5a_{n-3}$$

Here the subscripts present are $n-1$ and $n-3$. There is **no** a_{n-2} term at all.

- Largest subscript distance from n: 3 (from a_{n-3})
- Smallest subscript distance from n: 1 (from a_{n-1})
- Order $= 3 - 1 = 2$...

A *linear homogeneous recurrence relation of degree k with constant coefficients* is a recurrence relation of the form

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k},$$

where $c_1, c_2, \ldots, c_k$ are real numbers, and $c_k \neq 0$.

Constant coefficient

Definition 8.3: A linear recurrence equation has **constant coefficients** if coefficients of a_{n-1}, a_{n-2}, etc., are all constant, i.e., do not depend on the indices n, $n-1$, etc.

For example, the recurrence equation in Example 8.3 has constant coefficients, whereas the recurrence equation $a_n = na_{n-1} + 4$ does not satisfy the constant coefficient criterion.

homogeneous and non-homogeneous linear recurrence reln

Definition 8.4: A linear recurrence equation is called **homogeneous** if a_n is a linear function of past values a_{n-1}, a_{n-2}, etc., only and does not contain any other additional terms; otherwise, it is called **nonhomogeneous**.

Example 8.1 above is a nonhomogeneous linear recurrence equation and the recurrence equation given in Example 8.3 is a homogeneous linear recurrence equation.

A nonhomogeneous linear recurrence equation consists of two parts—the nonhomogeneous component $f(n)$ and the rest of the equation, called the homogeneous component. In Example 8.1, the homogeneous part is $a_n = 4a_{n-1}$ and the nonhomogeneous component is $f(n) = -2$.

general vs particular soln

Definition 8.5: The solution of a given linear recurrence equation is called the **general** solution. It consists of two parts, the first part obtained from the homogeneous part and the second part contributed by the nonhomogeneous component $f(n)$. This general solution must satisfy the given set of initial conditions. A solution so obtained in know as **the particular solution**.

Solving linear recurrence relations

Solving linear homogeneous recurrence relations with constant coefficients

We can use two key ideas to find all their solutions

all their solutions. First, these recurrence relations have solutions of the form $a_n = r^n$, where r is a constant. To see this, observe that $a_n = r^n$ is a solution of the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$ if and only if

$$r^n = c_1 r^{n-1} + c_2 r^{n-2} + \cdots + c_k r^{n-k}.$$

When both sides of this equation are divided by r^{n-k} (when $r \neq 0$) and the right-hand side is subtracted from the left, we obtain the equation

$$r^k - c_1 r^{k-1} - c_2 r^{k-2} - \cdots - c_{k-1} r - c_k = 0.$$

Consequently, the sequence $\{a_n\}$ with $a_n = r^n$ where $r \neq 0$ is a solution if and only if r is a solution of this last equation. We call this the **characteristic equation** of the recurrence relation.

Step 1: Divide Every Term by r^{n-k}

Since we assumed $r \neq 0$, we're allowed to divide both sides by r^{n-k}. Let's do this term by term, using the exponent rule $\frac{r^a}{r^b} = r^{a-b}$.

Left side:

$$\frac{r^n}{r^{n-k}} = r^{n-(n-k)} = r^k$$

Right side, term by term:

$$\frac{c_1 r^{n-1}}{r^{n-k}} = c_1 \cdot r^{(n-1)-(n-k)} = c_1 r^{k-1}$$

$$\frac{c_2 r^{n-2}}{r^{n-k}} = c_2 \cdot r^{(n-2)-(n-k)} = c_2 r^{k-2}$$

$$\vdots$$

$$\frac{c_{k-1} r^{n-(k-1)}}{r^{n-k}} = c_{k-1} \cdot r^{[n-(k-1)]-(n-k)} = c_{k-1} r^1$$

$$\frac{c_k r^{n-k}}{r^{n-k}} = c_k \cdot r^0 = c_k$$

The solutions of this equation are called the **characteristic roots** of the recurrence relation.

Linear combination of two solutions is also a solution

The other key observation is that a linear combination of two solutions of a linear homogeneous recurrence relation is also a solution. To see this, suppose that s_n and t_n are both solutions of $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$. Then

$$s_n = c_1 s_{n-1} + c_2 s_{n-2} + \cdots + c_k s_{n-k}$$

and

$$t_n = c_1 t_{n-1} + c_2 t_{n-2} + \cdots + c_k t_{n-k}.$$

Now suppose that b_1 and b_2 are real numbers. Then

$$\begin{aligned} b_1 s_n + b_2 t_n &= b_1(c_1 s_{n-1} + c_2 s_{n-2} + \cdots + c_k s_{n-k}) + b_2(c_1 t_{n-1} + c_2 t_{n-2} + \cdots + c_k t_{n-k}) \\ &= c_1(b_1 s_{n-1} + b_2 t_{n-1}) + c_2(b_1 s_{n-2} + b_2 t_{n-2}) + \cdots + c_k(b_1 s_{n-k} + b_k t_{n-k}). \end{aligned}$$

This means that $b_1 s_n + b_2 t_n$ is also a solution of the same linear homogeneous recurrence relation.

Solving linear recurrence relation of degree two

THEOREM 1 Let c_1 and c_2 be real numbers. Suppose that $r^2 - c_1 r - c_2 = 0$ has two distinct roots r_1 and r_2. Then the sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ if and only if $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$ for $n = 0, 1, 2, \ldots$, where α_1 and α_2 are constants.

Proof: We must do two things to prove the theorem. First, it must be shown that if r_1 and r_2 are the roots of the characteristic equation, and α_1 and α_2 are constants, then the sequence $\{a_n\}$ with $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$ is a solution of the recurrence relation. Second, it must be shown that if the sequence $\{a_n\}$ is a solution, then $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$ for some constants α_1 and α_2.

We now show that if $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$, then the sequence $\{a_n\}$ is a solution of the recurrence relation. Because r_1 and r_2 are roots of $r^2 - c_1 r - c_2 = 0$, it follows that $r_1^2 = c_1 r_1 + c_2$ and $r_2^2 = c_1 r_2 + c_2$.

From these equations, we see that

$$\begin{aligned} c_1 a_{n-1} + c_2 a_{n-2} &= c_1(\alpha_1 r_1^{n-1} + \alpha_2 r_2^{n-1}) + c_2(\alpha_1 r_1^{n-2} + \alpha_2 r_2^{n-2}) \\ &= \alpha_1 r_1^{n-2}(c_1 r_1 + c_2) + \alpha_2 r_2^{n-2}(c_1 r_2 + c_2) \\ &= \alpha_1 r_1^{n-2} r_1^2 + \alpha_2 r_2^{n-2} r_2^2 \\ &= \alpha_1 r_1^n + \alpha_2 r_2^n \\ &= a_n. \end{aligned}$$

This shows that the sequence $\{a_n\}$ with $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$ is a solution of the recurrence relation.

To show that every solution $\{a_n\}$ of the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ has $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$ for $n = 0, 1, 2, \ldots$, for some constants α_1 and α_2, suppose that $\{a_n\}$ is a solution of the recurrence relation, and the initial conditions $a_0 = C_0$ and $a_1 = C_1$ hold. It will be shown that there are constants α_1 and α_2 such that the sequence $\{a_n\}$ with $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$ satisfies these same initial conditions. This requires that

$$\begin{aligned} a_0 &= C_0 = \alpha_1 + \alpha_2, \\ a_1 &= C_1 = \alpha_1 r_1 + \alpha_2 r_2. \end{aligned}$$

We can solve these two equations for α_1 and α_2. From the first equation it follows that $\alpha_2 = C_0 - \alpha_1$. Inserting this expression into the second equation gives

$$C_1 = \alpha_1 r_1 + (C_0 - \alpha_1) r_2.$$

Hence,

$$C_1 = \alpha_1 (r_1 - r_2) + C_0 r_2.$$

This shows that

$$\alpha_1 = \frac{C_1 - C_0 r_2}{r_1 - r_2}$$

and

$$\alpha_2 = C_0 - \alpha_1 = C_0 - \frac{C_1 - C_0 r_2}{r_1 - r_2} = \frac{C_0 r_1 - C_1}{r_1 - r_2},$$

where these expressions for α_1 and α_2 depend on the fact that $r_1 \neq r_2$. (When $r_1 = r_2$, this theorem is not true.) Hence, with these values for α_1 and α_2, the sequence $\{a_n\}$ with $\alpha_1 r_1^n + \alpha_2 r_2^n$ satisfies the two initial conditions.

We know that $\{a_n\}$ and $\{\alpha_1 r_1^n + \alpha_2 r_2^n\}$ are both solutions of the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ and both satisfy the initial conditions when $n = 0$ and $n = 1$. Because there is a unique solution of a linear homogeneous recurrence relation of degree two with two initial conditions, it follows that the two solutions are the same, that is, $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$ for all nonnegative integers n. We have completed the proof by showing that a solution of the linear homogeneous recurrence relation with constant coefficients of degree two must be of the form $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$, where α_1 and α_2 are constants. ◁

EXAMPLE 3 What is the solution of the recurrence relation

Extra Examples

$$a_n = a_{n-1} + 2a_{n-2}$$

with $a_0 = 2$ and $a_1 = 7$?

Solution: Theorem 1 can be used to solve this problem. The characteristic equation of the recurrence relation is $r^2 - r - 2 = 0$. Its roots are $r = 2$ and $r = -1$. Hence, the sequence $\{a_n\}$ is a solution to the recurrence relation if and only if

$$a_n = \alpha_1 2^n + \alpha_2 (-1)^n,$$

for some constants α_1 and α_2. From the initial conditions, it follows that

$$a_0 = 2 = \alpha_1 + \alpha_2,$$
$$a_1 = 7 = \alpha_1 \cdot 2 + \alpha_2 \cdot (-1).$$

Solving these two equations shows that $\alpha_1 = 3$ and $\alpha_2 = -1$. Hence, the solution to the recurrence relation and initial conditions is the sequence $\{a_n\}$ with

$$a_n = 3 \cdot 2^n - (-1)^n.$$ ◀

Solving for fibonacci sequence.

Let me guess what's happening: Fibonacci's characteristic equation doesn't have nice integer roots like the example (2 and -1) — it has **irrational roots**, and solving for the constants involves messy algebra with square roots. That's usually where people get stuck or think they've made an error when actually they haven't. Let's do it fully, step by step.

Setup

$$a_n = a_{n-1} + a_{n-2}, \quad a_0 = 0,\ a_1 = 1$$

Step 1: Characteristic Equation

Following the same pattern as your example: rewrite as $a_n - a_{n-1} - a_{n-2} = 0$, substitute $a_n = r^n$:

$$r^2 - r - 1 = 0$$

Step 2: Find the Roots — Here's the First Difference

Unlike your example (which factored nicely into $r = 2, r = -1$), this one doesn't factor over the integers. You need the **quadratic formula**:

$$r = \frac{1 \pm \sqrt{1^2 - 4(1)(-1)}}{2} = \frac{1 \pm \sqrt{5}}{2}$$

So the two roots are:

$$r_1 = \frac{1 + \sqrt{5}}{2} \approx 1.618 \quad \text{(this is } \varphi\text{, the golden ratio)}$$

$$r_2 = \frac{1-\sqrt{5}}{2} \approx -0.618$$

This is likely where things "stop working" for you — if you were expecting clean integer roots like the worked example, irrational roots feel like a sign of error. They're not; Fibonacci genuinely has irrational characteristic roots. This is fine — the method still works exactly the same way, just with messier numbers.

Step 3: General Form

$$a_n = \alpha_1 \left(\frac{1+\sqrt{5}}{2}\right)^n + \alpha_2 \left(\frac{1-\sqrt{5}}{2}\right)^n$$

Step 4: Use Initial Conditions to Solve for α_1, α_2

At $n = 0$:

$$a_0 = 0 = \alpha_1 + \alpha_2 \quad \Rightarrow \quad \alpha_2 = -\alpha_1$$

At $n = 1$:

$$a_1 = 1 = \alpha_1 \left(\frac{1+\sqrt{5}}{2}\right) + \alpha_2 \left(\frac{1-\sqrt{5}}{2}\right)$$

Substitute $\alpha_2 = -\alpha_1$:

$$1 = \alpha_1 \left(\frac{1+\sqrt{5}}{2}\right) - \alpha_1 \left(\frac{1-\sqrt{5}}{2}\right)$$

$$1 = \alpha_1 \left[\frac{(1+\sqrt{5}) - (1-\sqrt{5})}{2}\right] = \alpha_1 \left[\frac{2\sqrt{5}}{2}\right] = \alpha_1 \sqrt{5}$$

So:

$$\alpha_1 = \frac{1}{\sqrt{5}}, \qquad \alpha_2 = -\frac{1}{\sqrt{5}}$$

This step is the second likely sticking point — the algebra involves combining fractions with $\sqrt{5}$ in them, and it's easy to drop a sign or mismanage the fraction. Go slowly through that bracket simplification if you're redoing it by hand.

Step 5: Final Closed Form — Binet's Formula

$$a_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2}\right)^n - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2}\right)^n$$

$$\boxed{a_n = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right]}$$

This is the famous **Binet's formula**.

Sanity Check

Let's verify a_2 should be 1 (since $0, 1, 1, 2, 3, 5, \ldots$):

$$a_2 = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^2 - \left(\frac{1-\sqrt{5}}{2}\right)^2\right]$$

$\left(\frac{1+\sqrt{5}}{2}\right)^2 = \frac{1+2\sqrt{5}+5}{4} = \frac{6+2\sqrt{5}}{4} = \frac{3+\sqrt{5}}{2}$

$\left(\frac{1-\sqrt{5}}{2}\right)^2 = \frac{1-2\sqrt{5}+5}{4} = \frac{6-2\sqrt{5}}{4} = \frac{3-\sqrt{5}}{2}$

Difference: $\frac{3+\sqrt{5}}{2} - \frac{3-\sqrt{5}}{2} = \frac{2\sqrt{5}}{2} = \sqrt{5}$

So $a_2 = \frac{1}{\sqrt{5}} \cdot \sqrt{5} = 1$ ✓

It checks out — and remarkably, even though the formula is stuffed with $\sqrt{5}$'s, every output a_n comes out as a whole number, because the irrational parts always cancel.

Exercise

$$a_n = 5a_{n-1} - 6a_{n-2}, \quad a_0 = 1,\ a_1 = 4$$

THEOREM 2

Let c_1 and c_2 be real numbers with $c_2 \neq 0$. Suppose that $r^2 - c_1 r - c_2 = 0$ has only one root r_0. A sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ if and only if $a_n = \alpha_1 r_0^n + \alpha_2 n r_0^n$, for $n = 0, 1, 2, \ldots$, where α_1 and α_2 are constants.

EXAMPLE 5 What is the solution of the recurrence relation

$$a_n = 6a_{n-1} - 9a_{n-2}$$

with initial conditions $a_0 = 1$ and $a_1 = 6$?

Solution: The only root of $r^2 - 6r + 9 = 0$ is $r = 3$. Hence, the solution to this recurrence relation is

$$a_n = \alpha_1 3^n + \alpha_2 n 3^n$$

for some constants α_1 and α_2. Using the initial conditions, it follows that

$$a_0 = 1 = \alpha_1,$$
$$a_1 = 6 = \alpha_1 \cdot 3 + \alpha_2 \cdot 3.$$

Solving these two equations shows that $\alpha_1 = 1$ and $\alpha_2 = 1$. Consequently, the solution to this recurrence relation and the initial conditions is

$$a_n = 3^n + n3^n.$$

◀

THEOREM 3 Let $c_1, c_2, \ldots, c_k$ be real numbers. Suppose that the characteristic equation

$$r^k - c_1 r^{k-1} - \cdots - c_k = 0$$

has k distinct roots $r_1, r_2, \ldots, r_k$. Then a sequence $\{a_n\}$ is a solution of the recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$$

if and only if

$$a_n = \alpha_1 r_1^n + \alpha_2 r_2^n + \cdots + \alpha_k r_k^n$$

for $n = 0, 1, 2, \ldots$, where $\alpha_1, \alpha_2, \ldots, \alpha_k$ are constants.

EXAMPLE 6 Find the solution to the recurrence relation

$$a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$$

with the initial conditions $a_0 = 2$, $a_1 = 5$, and $a_2 = 15$.

Solution: The characteristic polynomial of this recurrence relation is

$$r^3 - 6r^2 + 11r - 6.$$

The characteristic roots are $r = 1$, $r = 2$, and $r = 3$, because $r^3 - 6r^2 + 11r - 6 = (r - 1)(r - 2)(r - 3)$. Hence, the solutions to this recurrence relation are of the form

$$a_n = \alpha_1 \cdot 1^n + \alpha_2 \cdot 2^n + \alpha_3 \cdot 3^n.$$

To find the constants α_1, α_2, and α_3, use the initial conditions. This gives

$$\begin{aligned} a_0 &= 2 = \alpha_1 + \alpha_2 + \alpha_3, \\ a_1 &= 5 = \alpha_1 + \alpha_2 \cdot 2 + \alpha_3 \cdot 3, \\ a_2 &= 15 = \alpha_1 + \alpha_2 \cdot 4 + \alpha_3 \cdot 9. \end{aligned}$$

When these three simultaneous equations are solved for α_1, α_2, and α_3, we find that $\alpha_1 = 1$, $\alpha_2 = -1$, and $\alpha_3 = 2$. Hence, the unique solution to this recurrence relation and the given initial conditions is the sequence $\{a_n\}$ with

$$a_n = 1 - 2^n + 2 \cdot 3^n.$$ ◀

THEOREM 4 Let $c_1, c_2, \ldots, c_k$ be real numbers. Suppose that the characteristic equation

$$r^k - c_1 r^{k-1} - \cdots - c_k = 0$$

has t distinct roots $r_1, r_2, \ldots, r_t$ with multiplicities $m_1, m_2, \ldots, m_t$, respectively, so that $m_i \geq 1$ for $i = 1, 2, \ldots, t$ and $m_1 + m_2 + \cdots + m_t = k$. Then a sequence $\{a_n\}$ is a solution of the recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$$

if and only if

$$\begin{aligned} a_n = {} & (\alpha_{1,0} + \alpha_{1,1} n + \cdots + \alpha_{1,m_1-1} n^{m_1-1}) r_1^n \\ & + (\alpha_{2,0} + \alpha_{2,1} n + \cdots + \alpha_{2,m_2-1} n^{m_2-1}) r_2^n \\ & + \cdots + (\alpha_{t,0} + \alpha_{t,1} n + \cdots + \alpha_{t,m_t-1} n^{m_t-1}) r_t^n \end{aligned}$$

for $n = 0, 1, 2, \ldots$, where $\alpha_{i,j}$ are constants for $1 \leq i \leq t$ and $0 \leq j \leq m_i - 1$.

EXAMPLE 7 Suppose that the roots of the characteristic equation of a linear homogeneous recurrence relation are 2, 2, 2, 5, 5, and 9 (that is, there are three roots, the root 2 with multiplicity three, the root 5 with multiplicity two, and the root 9 with multiplicity one). What is the form of the general solution?

Solution: By Theorem 4, the general form of the solution is

$$(\alpha_{1,0} + \alpha_{1,1} n + \alpha_{1,2} n^2) 2^n + (\alpha_{2,0} + \alpha_{2,1} n) 5^n + \alpha_{3,0} 9^n.$$ ◀

EXAMPLE 8 Find the solution to the recurrence relation

$$a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3}$$

with initial conditions $a_0 = 1$, $a_1 = -2$, and $a_2 = -1$.

Solution: The characteristic equation of this recurrence relation is

$$r^3 + 3r^2 + 3r + 1 = 0.$$

Because $r^3 + 3r^2 + 3r + 1 = (r + 1)^3$, there is a single root $r = -1$ of multiplicity three of the characteristic equation. By Theorem 4 the solutions of this recurrence relation are of the form

$$a_n = \alpha_{1,0}(-1)^n + \alpha_{1,1}n(-1)^n + \alpha_{1,2}n^2(-1)^n.$$

To find the constants $\alpha_{1,0}$, $\alpha_{1,1}$, and $\alpha_{1,2}$, use the initial conditions. This gives

$$a_0 = 1 = \alpha_{1,0},$$
$$a_1 = -2 = -\alpha_{1,0} - \alpha_{1,1} - \alpha_{1,2},$$
$$a_2 = -1 = \alpha_{1,0} + 2\alpha_{1,1} + 4\alpha_{1,2}.$$

The simultaneous solution of these three equations is $\alpha_{1,0} = 1$, $\alpha_{1,1} = 3$, and $\alpha_{1,2} = -2$. Hence, the unique solution to this recurrence relation and the given initial conditions is the sequence $\{a_n\}$ with

$$a_n = (1 + 3n - 2n^2)(-1)^n.$$

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Solving Recurrence Relations by Substitution (Iteration Method)

What This Method Is

The **substitution method** (also called the **iteration method**) solves a recurrence by repeatedly substituting the recurrence relation into itself, peeling back one layer at a time, until a visible pattern emerges. You then express that pattern as a closed-form formula and verify it — often using induction for full rigor.

This is the most elementary of the three techniques on your syllabus (substitution, characteristic equations, generating functions). It builds direct intuition for *why* a closed form looks the way it does, before moving to the more mechanical algebraic methods.

Best suited for: order-1 recurrences, and recurrences with a simple, regular structure. It gets unwieldy fast for order-2+ or irregular recurrences — which is exactly the motivation for introducing characteristic equations and generating functions afterward.

The General Procedure

1. Write the recurrence for a_n in terms of a_{n-1}.
2. Replace a_{n-1} using the same recurrence (shifted down by one index), giving a_n in terms of a_{n-2}.
3. Repeat this substitution again and again.
4. After a few rounds, a **pattern** appears in terms of how many times you've substituted.
5. Generalize the pattern to the point where you hit the known **initial condition** (usually a_0).
6. Write the closed form, then **verify** by checking it against small values, or by direct substitution back into the original recurrence, or by induction.

Example 1: Simple Multiplicative Recurrence

$$a_n = 3a_{n-1}, \qquad a_0 = 2$$

Unroll step by step:

$$a_n = 3a_{n-1}$$

$$= 3(3a_{n-2}) = 3^2 a_{n-2}$$

$$= 3^2(3a_{n-3}) = 3^3 a_{n-3}$$

$$= 3^3(3a_{n-4}) = 3^4 a_{n-4}$$

Spot the pattern: after k substitutions, $a_n = 3^k a_{n-k}$.

Push to the initial condition: we want $n - k = 0$, i.e. $k = n$:

$$a_n = 3^n a_0 = 3^n \cdot 2 = 2 \cdot 3^n$$

Verify: Check $n = 2$ directly from the recurrence: $a_1 = 3(2) = 6$, $a_2 = 3(6) = 18$. Formula: $2 \cdot 3^2 = 18$ ✓

Example 2: Additive Recurrence

$$a_n = a_{n-1} + 3, \qquad a_0 = 2$$

Unroll:

$$a_n = a_{n-1} + 3$$

$$= (a_{n-2} + 3) + 3 = a_{n-2} + 2(3)$$

$$= (a_{n-3} + 3) + 2(3) = a_{n-3} + 3(3)$$

$$= a_{n-4} + 4(3)$$

Pattern: after k substitutions, $a_n = a_{n-k} + k(3)$.

Push to $k = n$:

$$a_n = a_0 + 3n = 2 + 3n$$

Verify: a_1 should be $2 + 3 = 5$. Directly: $a_1 = a_0 + 3 = 2 + 3 = 5$ ✓. $a_2 = 2 + 6 = 8$; directly $a_2 = a_1 + 3 = 5 + 3 = 8$ ✓

Example 3: Combined Multiplicative + Additive (a step up in difficulty)

$$a_n = 2a_{n-1} + 1, \qquad a_0 = 1$$

Unroll carefully — track both the coefficient and the constant:

$$a_n = 2a_{n-1} + 1$$

$$= 2(2a_{n-2} + 1) + 1 = 4a_{n-2} + 2 + 1 = 2^2 a_{n-2} + (2 + 1)$$

$$= 2^2(2a_{n-3} + 1) + (2 + 1) = 2^3 a_{n-3} + (4 + 2 + 1)$$

$$= 2^3(2a_{n-4} + 1) + (4 + 2 + 1) = 2^4 a_{n-4} + (8 + 4 + 2 + 1)$$

Pattern after k substitutions:

$$a_n = 2^k a_{n-k} + \left(2^{k-1} + 2^{k-2} + \cdots + 2 + 1\right) = 2^k a_{n-k} + (2^k - 1)$$

using the geometric series sum $2^{k-1} + \cdots + 1 = 2^k - 1$.

Push to $k = n$:

$$a_n = 2^n a_0 + (2^n - 1) = 2^n(1) + 2^n - 1 = 2^{n+1} - 1$$

Verify: a_1 should be $2(1) + 1 = 3$. Formula: $2^2 - 1 = 3$ ✓. $a_2 = 2(3) + 1 = 7$; formula: $2^3 - 1 = 7$ ✓

Teaching note: this example is worth including precisely *because* it requires recognizing a geometric series mid-derivation — a good bridge skill before generating functions, where sums like this appear constantly. It also matches the non-homogeneous generating-function example from earlier notes ($a_n = 2a_{n-1} + 1$), so you can show students the substitution method and the GF method converging on the exact same answer, $a_n = 2^{n+1} - 1$.

Example 4: A Case Where Substitution Gets Messy (motivating the next method)

$$a_n = a_{n-1} + a_{n-2}, \qquad a_0 = 0,\ a_1 = 1 \quad \text{(Fibonacci)}$$

Try unrolling this the same way:

$$a_n = a_{n-1} + a_{n-2}$$

$$= (a_{n-2} + a_{n-3}) + a_{n-2} = 2a_{n-2} + a_{n-3}$$

$$= 2(a_{n-3} + a_{n-4}) + a_{n-3} = 3a_{n-3} + 2a_{n-4}$$

$$= 3(a_{n-4} + a_{n-5}) + 2a_{n-4} = 5a_{n-4} + 3a_{n-5}$$

No clean closed-form pattern emerges — the coefficients (1, 1, 2, 3, 5, ...) are themselves Fibonacci numbers, so we're going in circles. This is the key teaching moment: substitution works beautifully for order-1 (and some very regular order-2) recurrences, but for something like Fibonacci, it breaks down completely, and you *need* the characteristic equation or generating function method instead.

This is a great line to say explicitly in class: "Substitution is our simplest tool, but it's not universal — watch what happens when we try it on Fibonacci" — then pivot directly into the characteristic-equation lecture, since students will have just watched the method fail in real time.

Verifying by Induction (the rigorous finish)

Once you've *guessed* a closed form via substitution, it's good practice (and often required for full marks) to **prove it by mathematical induction**, since the substitution process itself is really "pattern recognition," not a proof.

Example, finishing off Example 2 ($a_n = a_{n-1} + 3$, $a_0 = 2$, claimed closed form $a_n = 2 + 3n$):

Base case: $a_0 = 2 + 3(0) = 2$ ✓ matches given initial condition.

Inductive step: Assume $a_{n-1} = 2 + 3(n-1)$ holds. Show $a_n = 2 + 3n$ follows:

$$a_n = a_{n-1} + 3 = [2 + 3(n-1)] + 3 = 2 + 3n - 3 + 3 = 2 + 3n \checkmark$$

This closes the loop rigorously.

Summary for Students

Step	What to do
1	Substitute the recurrence into itself repeatedly
2	Track the pattern in exponents/coefficients as you go
3	Express the pattern after k substitutions
4	Set k so the recursion bottoms out at a known initial value
5	Simplify (watch for geometric series sums)
6	Verify — by direct check on small n, or full induction proof

Where this fits in your syllabus flow: teach substitution first (Examples 1–3) to build comfort with "unrolling," then use Example 4 (Fibonacci breaking down) as the natural transition into characteristic equations and generating functions, both of which handle order-2+ recurrences that substitution can't crack cleanly.z

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Here's a classwork set — same structure as Examples 2 and 3, with increasing difficulty. I'd suggest giving students the problems first without answers, then providing the answer key separately (either after class or on a following slide).

Type 2 Classwork: Additive Recurrences ($a_n = a_{n-1} + f(n)$ or $+c$)

CW2.1 (straightforward — same shape as the worked example)

$$a_n = a_{n-1} + 5, \qquad a_0 = 3$$

CW2.2 (different constant, different start — checks they're not just pattern-matching numbers)

$$a_n = a_{n-1} - 2, \qquad a_0 = 10$$

CW2.3 (slightly harder — the added term depends on n, not constant)

$$a_n = a_{n-1} + n, \qquad a_0 = 0$$

(This one requires summing $1 + 2 + \cdots + n$ instead of a constant times n — good stretch problem.)

CW2.4 (added term is $2n$ — tests whether they can generalize CW2.3)

$$a_n = a_{n-1} + 2n, \qquad a_0 = 1$$

CW2.5 (added term is n^2 — real stretch, needs the sum-of-squares formula)

$$a_n = a_{n-1} + n^2, \qquad a_0 = 0$$

Answer Key — Type 2

CW2.1: $a_n = 3 + 5n$ Check: $a_1 = 8$; formula gives $3 + 5 = 8$ ✓

CW2.2: $a_n = 10 - 2n$ Check: $a_1 = 8$; formula: $10 - 2 = 8$ ✓

CW2.3: Unrolling gives $a_n = a_0 + (1 + 2 + \cdots + n) = 0 + \dfrac{n(n+1)}{2}$

$$a_n = \frac{n(n+1)}{2}$$

Check: a_3 directly: $a_1 = 1, a_2 = 3, a_3 = 6$. Formula: $\frac{3 \cdot 4}{2} = 6$ ✓

CW2.4: $a_n = 1 + 2(1 + 2 + \cdots + n) = 1 + 2 \cdot \dfrac{n(n+1)}{2} = 1 + n(n+1)$

$$a_n = n^2 + n + 1$$

Check: a_2 directly: $a_1 = 1 + 2 = 3, a_2 = 3 + 4 = 7$. Formula: $4 + 2 + 1 = 7$ ✓

CW2.5: $a_n = 0 + (1^2 + 2^2 + \cdots + n^2) = \dfrac{n(n+1)(2n+1)}{6}$

$$a_n = \frac{n(n+1)(2n+1)}{6}$$

Check: a_2 directly: $a_1 = 1, a_2 = 1 + 4 = 5$. Formula: $\frac{2 \cdot 3 \cdot 5}{6} = 5$ ✓

Type 3 Classwork: Multiplicative + Additive Recurrences ($a_n = c, a_{n-1} + d$)

CW3.1 (same coefficient as the worked example, different constants — direct practice)

$$a_n = 2a_{n-1} + 3, \qquad a_0 = 1$$

CW3.2 (different multiplier — tests generalization of the geometric series step)

$$a_n = 3a_{n-1} + 2, \qquad a_0 = 1$$

CW3.3 (negative additive constant — watch sign handling)

$$a_n = 2a_{n-1} - 1, \qquad a_0 = 2$$

CW3.4 (larger multiplier, small constant — heavier arithmetic, same method)

$$a_n = 4a_{n-1} + 3, \qquad a_0 = 0$$

CW3.5 (challenge — multiplier is not an integer, tests whether they trust the algebra over "nice number" instincts, callback to the Fibonacci/fractions

discussion)

$$a_n = \tfrac{1}{2}a_{n-1} + 1, \qquad a_0 = 0$$

Answer Key — Type 3

General pattern to remind them of: for $a_n = c, a_{n-1} + d$ (with $c \neq 1$), unrolling gives

$$a_n = c^n a_0 + d(c^{n-1} + c^{n-2} + \cdots + 1) = c^n a_0 + d \cdot \frac{c^n - 1}{c - 1}$$

CW3.1: $c = 2, d = 3, a_0 = 1$

$$a_n = 2^n(1) + 3 \cdot \frac{2^n - 1}{2 - 1} = 2^n + 3(2^n - 1) = 4 \cdot 2^n - 3$$

Check: $a_1 = 2(1) + 3 = 5$; formula: $4(2) - 3 = 5$ ✓

CW3.2: $c = 3, d = 2, a_0 = 1$

$$a_n = 3^n(1) + 2 \cdot \frac{3^n - 1}{3 - 1} = 3^n + (3^n - 1) = 2 \cdot 3^n - 1$$

Check: $a_1 = 3(1) + 2 = 5$; formula: $2(3) - 1 = 5$ ✓

CW3.3: $c = 2, d = -1, a_0 = 2$

$$a_n = 2^n(2) + (-1) \cdot \frac{2^n - 1}{2 - 1} = 2^{n+1} - (2^n - 1) = 2^{n+1} - 2^n + 1 = 2^n + 1$$

Check: $a_1 = 2(2) - 1 = 3$; formula: $2^1 + 1 = 3$ ✓

CW3.4: $c = 4, d = 3, a_0 = 0$

$$a_n = 4^n(0) + 3 \cdot \frac{4^n - 1}{4 - 1} = 4^n - 1$$

Check: $a_1 = 4(0) + 3 = 3$; formula: $4^1 - 1 = 3$ ✓

CW3.5: $c = \frac{1}{2}, d = 1, a_0 = 0$

$$a_n = \left(\tfrac{1}{2}\right)^n(0) + 1 \cdot \frac{(\frac{1}{2})^n - 1}{\frac{1}{2} - 1} = \frac{(\frac{1}{2})^n - 1}{-\frac{1}{2}} = 2\left[1 - \left(\tfrac{1}{2}\right)^n\right]$$

Check: $a_1 = \frac{1}{2}(0) + 1 = 1$; formula: $2[1 - \frac{1}{2}] = 2(\frac{1}{2}) = 1$ ✓

Worth pointing out for CW3.5: as $n \to \infty$, $a_n \to 2$ — the sequence converges rather than blowing up, since $|c| < 1$. Nice discussion point if you want to briefly touch on stability/convergence of recurrences, even though that's likely beyond this unit's main scope.

Suggested Classwork Delivery

- Give **CW2.1–CW2.2** and **CW3.1–CW3.3** as timed in-class work (5–7 min) — these mirror the worked examples closely enough that students should succeed independently.
- Reserve **CW2.3–2.5** and **CW3.4–3.5** as either bonus/stretch problems in class or take-home homework, since they require the extra step of recognizing a sum formula (arithmetic series, sum of squares) or handling non-integer coefficients.

Want me to also prepare a version of these as a printable worksheet (no answers shown), separate from this answer-key version?

Here's the full step-by-step unrolling for **CW2.3**:

$$a_n = a_{n-1} + n, \qquad a_0 = 0$$

Step-by-Step Substitution

$$a_n = a_{n-1} + n$$

Substitute $a_{n-1} = a_{n-2} + (n-1)$:

$$a_n = [a_{n-2} + (n-1)] + n = a_{n-2} + (n-1) + n$$

Substitute $a_{n-2} = a_{n-3} + (n-2)$:

$$a_n = [a_{n-3} + (n-2)] + (n-1) + n = a_{n-3} + (n-2) + (n-1) + n$$

Substitute $a_{n-3} = a_{n-4} + (n-3)$:

$$a_n = a_{n-4} + (n-3) + (n-2) + (n-1) + n$$

Spot the Pattern

Notice what's accumulating on the right: after k substitutions, we have

$$a_n = a_{n-k} + \Big[(n-k+1) + (n-k+2) + \cdots + (n-1) + n\Big]$$

That bracketed sum is just consecutive integers running from $n-k+1$ up to n.

Push to the Initial Condition

We want to unroll until we hit a_0, i.e., set $n - k = 0 \implies k = n$:

$$a_n = a_0 + \Big[1 + 2 + 3 + \cdots + n\Big]$$

The term in brackets is now the sum of **all** integers from 1 to n — this is exactly the same sum whether you build it up from the top (as we did) or list it starting from 1; it's the same set of numbers either way.

Apply the Known Sum Formula

Using the standard arithmetic series formula:

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

And since $a_0 = 0$:

$$\boxed{a_n = \frac{n(n+1)}{2}}$$

Verification

Direct computation from the recurrence:

- $a_1 = a_0 + 1 = 0 + 1 = 1$
- $a_2 = a_1 + 2 = 1 + 2 = 3$
- $a_3 = a_2 + 3 = 3 + 3 = 6$
- $a_4 = a_3 + 4 = 6 + 4 = 10$

Formula check:

- $a_1 = \frac{1 \cdot 2}{2} = 1$ ✓
- $a_2 = \frac{2 \cdot 3}{2} = 3$ ✓
- $a_3 = \frac{3 \cdot 4}{2} = 6$ ✓
- $a_4 = \frac{4 \cdot 5}{2} = 10$ ✓

All match.

Nice Point to Make to Students

This closed form, $\frac{n(n+1)}{2}$, is the **triangular numbers** formula — worth mentioning if students have seen it before (e.g., "sum of first n natural numbers," Gauss's classic trick of pairing first-and-last terms). It's a good moment to connect this recurrence-solving exercise back to a formula they likely already know from a different context, which reinforces that recurrences are just another lens on sequences they've already encountered.

Solution: $a_n = a_{n-1} + n$, with $a_0 = 0$

Step 1: Unroll the Recurrence by Repeated Substitution

Start with the recurrence itself:

$$a_n = a_{n-1} + n$$

Substitute $a_{n-1} = a_{n-2} + (n-1)$:

$$a_n = a_{n-2} + (n-1) + n$$

Substitute $a_{n-2} = a_{n-3} + (n-2)$:

$$a_n = a_{n-3} + (n-2) + (n-1) + n$$

Substitute $a_{n-3} = a_{n-4} + (n-3)$:

$$a_n = a_{n-4} + (n-3) + (n-2) + (n-1) + n$$

Step 2: Generalize to k Substitutions

The pattern after k substitutions is:

$$a_n = a_{n-k} + \Big[(n-k+1) + (n-k+2) + \cdots + (n-1) + n\Big]$$

Step 3: Push to the Initial Condition

We know a_0, so set $n - k = 0$, i.e., $k = n$:

$$a_n = a_0 + \Big[1 + 2 + 3 + \cdots + n\Big]$$

Step 4: Apply the Arithmetic Series Formula

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

Since $a_0 = 0$:

$$\boxed{a_n = \frac{n(n+1)}{2}}$$

Step 5: Verification

Direct computation from the recurrence:

$$a_0 = 0, \quad a_1 = 1, \quad a_2 = 3, \quad a_3 = 6, \quad a_4 = 10$$

Formula check:

$$a_1 = \tfrac{1\cdot 2}{2} = 1\checkmark \qquad a_2 = \tfrac{2\cdot 3}{2} = 3\checkmark \qquad a_3 = \tfrac{3\cdot 4}{2} = 6\checkmark \qquad a_4 = \tfrac{4\cdot 5}{2} = 10\checkmark$$

All values match — this is the **triangular numbers** sequence, $a_n = \dfrac{n(n+1)}{2}$.

Generating Functions

Definition 8.6: Given a sequence of numbers $a_0, a_1, \ldots, a_n, \ldots$ the associated generating function is:

$$A(z) = a_0 + a_1 z + a_2 z^2 + \ldots + a_n z^n + \ldots,$$

where z is a variable.

Let $A(z) = \sum_{n\geq 0} a_n z^n$, $B(z) = \sum_{n\geq 0} b_n z^n$, and $C(z) = \sum_{n\geq 0} c_n z^n$. We have the following properties:

1. $A(z) + B(z) = C(z)$ if and only if $\forall n \geq 0\ [a_n + b_n = c_n]$.
2. $A(z) \times B(z) = C(z)$ if and only if $\forall n \geq 0\ [c_n = \sum_{i=0}^{n} a_i \cdot b_{n-i}]$.

Some Useful Power Series and Their Closed Forms

In order to effectively use the generation functions, it is important to know some important sums. The following is a list of some useful power series and their closed forms. One may refer to the calculus textbooks for their proofs.

Let r be any real number and n be ant natural numbers.

$$\begin{aligned}
\frac{1}{1-sz} &= \sum_{0\leq i} s^i z^i \\
\frac{1}{(1-sz)^2} &= \sum_{0\leq i} (i+1) s^i z^i \\
(1+sz)^n &= \sum_{0\leq i\leq n} \binom{n}{i} s^i z^i \\
\frac{1}{(1+sz)^n} &= \sum_{0\leq i} \binom{-n}{i} s^i z^i \\
\ln(1+sz) &= \sum_{1\leq i} \frac{(-1)^{i+1}}{i} s^i z^i \\
-\ln(1-sz) &= \sum_{1\leq i} \frac{1}{i} s^i z^i \\
e^{sz} &= \sum_{0\leq i} \frac{1}{i!} s^i z^i
\end{aligned}$$

Solving recurrence relation by generating functions

Using Generating Functions to Solve Recurrence Equations

In brief, the key steps and idea behind the use of a generating function to solve a recurrence equation are:

Step 1: Write down the generating function of the given sequence.

Step 2: Use the given relation between a_n and a_{n-1} etc. to remove the recurrence from the generating function and simplify it.

Step 3: Expand the generating function obtained to in powers of z.

Step 4: Equate the coefficients of z^n in two different expressions of the generating functions of the same sequence. Thus obtain the solution of the given recurrence equation.

Exercises

q1

Example 8.9 Consider the recurrence:

$$a_0 = 1, \text{ and for } n \geq 1, \ a_n = 2a_{n-1} + 1.$$

In the first step, all that we need to do is to write the generating function:

$$A(z) = a_0 + a_1 z + a_2 z^2 + \ldots + a_n z^n + \ldots.$$

Next, by looking at the given recurrence relation we realize that for all values of n occurrences of $a_n - 2a_{n-1}$ can be replaced by 1. To utilize this property we multiply $A(z)$ by $2z$ and subtract it from $A(z)$, i.e., perform the following manipulations.

$$\begin{array}{rl} A(z) = & a_0 + a_1 z + a_2 z^2 + a_3 z^3 + \cdots + a_n z^n + \cdots \\ 2zA(z) = & \quad + 2a_0 z + 2a_1 z^2 + 2a_3 z^3 + \cdots + 2a_{n-1} z^n + \cdots \\ \hline (1-2z)A(z) = & a_0 + 1z + 1z^2 + 1z^3 + \cdots + 1z^n + \cdots \end{array}$$

We have,

$$(1-2z)A(z) = 1 + z + z^2 + z^3 + \cdots = \frac{1}{1-z}$$

because $a_0 = 1$. Consequently,

$$A(z) = \frac{1}{(1-2z)(1-z)} = \frac{2}{1-2z} + \frac{-1}{1-z}.$$

We know that

$$\frac{2}{1-2z} = \sum_{0 \leq i} 2 \cdot 2^i z^i \text{ and } \frac{-1}{1-z} = \sum_{0 \leq i} (-1) \cdot z^i.$$

Thus,

$$A(z) = \sum_{0 \leq i} 2 \cdot 2^i z^i + \sum_{0 \leq i} (-1) \cdot z^i = \sum_{0 \leq i} (2^{i+1} - 1) z^i.$$

Therefore, $a_n = 2^{n+1} - 1$. □

Let's build $2zA(z)$ carefully from scratch — it's just algebra applied term-by-term to the series, but the index shift is the part that trips people up visually.

Starting Point

$$A(z) = a_0 + a_1 z + a_2 z^2 + a_3 z^3 + \cdots + a_n z^n + \cdots$$

We want $2zA(z)$ — multiply the *entire* series by $2z$.

Step 1: Multiply Every Term by $2z$

Do it term by term, using the rule $z \cdot z^k = z^{k+1}$ (multiplying by z bumps the exponent up by one):

$$2z \cdot a_0 = 2a_0 z$$

$$2z \cdot a_1 z = 2a_1 z^2$$

$$2z \cdot a_2 z^2 = 2a_2 z^3$$

$$2z \cdot a_3 z^3 = 2a_3 z^4$$

$$\vdots$$

$$2z \cdot a_n z^n = 2a_n z^{n+1}$$

Step 2: Add Them All Up

$$2zA(z) = 2a_0 z + 2a_1 z^2 + 2a_2 z^3 + 2a_3 z^4 + \cdots + 2a_n z^{n+1} + \cdots$$

Step 3: This Is a Series in Powers of z — Relabel to See It Clearly

Right now, every term's coefficient index is **one less** than its power of z (e.g., $2a_0$ sits with z^1, $2a_1$ sits with z^2, $2a_2$ sits with z^3). To line this up under $A(z)$ for the subtraction in the image, we want to express $2zA(z)$ as "coefficient times z^n" — i.e., relabel using a new index.

Let $m = n + 1$ (so wherever we see z^{n+1}, call it z^m, meaning the coefficient sitting at power z^m is $2a_{m-1}$):

$$2zA(z) = \sum_{m=1}^{\infty} 2a_{m-1} z^m$$

Renaming m back to n (it's just a dummy label, so this is legal):

$$2zA(z) = \sum_{n=1}^{\infty} 2a_{n-1} z^n = 2a_0 z + 2a_1 z^2 + 2a_2 z^3 + \cdots + 2a_{n-1} z^n + \cdots$$

This is exactly the second row in the image:

$$2zA(z) = \qquad 2a_0 z + 2a_1 z^2 + 2a_3 z^3 + \cdots + 2a_{n-1} z^n + \cdots$$

Why the Row Is Shifted Right by One Column

This is the key visual point in the image: notice $2zA(z)$ has **no constant term** (nothing sits under a_0's column), and every term is shifted one column to the right compared to $A(z)$. That's a direct consequence of multiplying by z: every power of z increased by exactly 1, so what was the "z^n coefficient" in $A(z)$ (namely a_n) becomes the "z^{n+1} coefficient" in $2zA(z)$ (namely $2a_n$) — or equivalently, the coefficient sitting at z^n in $2zA(z)$ is $2a_{n-1}$, one index behind.

Why This Setup Is Useful (the punchline)

This shift is *exactly* what's needed to line up with the recurrence $a_n = 2a_{n-1} + 1$. When you subtract $2zA(z)$ from $A(z)$ column-by-column (matching same powers of z), the coefficient of z^n becomes:

$$a_n - 2a_{n-1}$$

which by the recurrence equals exactly 1 — that's why every column (from z^1 onward) collapses to just "$1z^n$" in the bottom row of the image. The whole point of multiplying by $2z$ was to manufacture this exact alignment.

Let's derive this partial fraction decomposition step by step — this is the same technique from the earlier notes (Section 4, Method 2), just applied to this specific example.

Where This Comes From

Continuing Example 8.9: after dividing both sides by $(1-2z)$, you'd get

$$A(z) = \frac{1}{(1-2z)(1-z)}$$

(This follows from $(1-2z)A(z) = \frac{1}{1-z}$ — the closed form of the geometric series on the right side of the earlier image — then dividing both sides by $(1-2z)$.)

Step 1: Set Up the Partial Fraction Form

Since the denominator factors into two distinct linear factors, $(1-2z)$ and $(1-z)$, we assume:

$$\frac{1}{(1-2z)(1-z)} = \frac{A}{1-2z} + \frac{B}{1-z}$$

for some constants A and B to be determined.

Step 2: Clear the Denominators

Multiply both sides by $(1-2z)(1-z)$:

$$1 = A(1-z) + B(1-2z)$$

Step 3: Solve for A and B

This equation must hold for **all** values of z, so we can plug in convenient values to isolate each constant.

Find A: choose z so that the B term vanishes. Set $1-2z=0 \implies z=\frac{1}{2}$:

$$1 = A\left(1-\frac{1}{2}\right) + B(0) = A\cdot\frac{1}{2}$$

$$A = 2$$

Find B: choose z so that the A term vanishes. Set $1-z=0 \implies z=1$:

$$1 = A(0) + B(1-2(1)) = B(-1)$$

$$B = -1$$

Step 4: Write the Result

$$\frac{1}{(1-2z)(1-z)} = \frac{2}{1-2z} + \frac{-1}{1-z}$$

which is exactly what's shown in the image.

Verification (always good to show students)

Combine the right side back over a common denominator to confirm it matches the left:

$$\frac{2}{1-2z} - \frac{1}{1-z} = \frac{2(1-z)-(1-2z)}{(1-2z)(1-z)} = \frac{2-2z-1+2z}{(1-2z)(1-z)} = \frac{1}{(1-2z)(1-z)}$$

✓

The z terms cancel perfectly, leaving just 1 in the numerator — confirming the decomposition is correct.

Why This Step Matters (the payoff)

This decomposition is the whole reason we do partial fractions: each piece, $\frac{2}{1-2z}$ and $\frac{-1}{1-z}$, is now a **simple geometric series** we already know how to expand:

$$\frac{2}{1-2z} = 2\sum_{n=0}^{\infty}(2z)^n = \sum_{n=0}^{\infty} 2^{n+1}z^n$$

$$\frac{-1}{1-z} = -\sum_{n=0}^{\infty} z^n$$

Adding these coefficient-by-coefficient gives:

$$a_n = 2^{n+1} - 1$$

— which matches exactly the answer you got earlier via the substitution method for this same recurrence ($a_n = 2a_{n-1} + 1$)! A nice moment to point out to students: two completely different techniques (substitution and generating functions) converging on the identical closed form.

This last step is really just restating the definition of a generating function in reverse — reading the coefficient straight off the series.

Recall What $A(z)$ Means in the First Place

Right at the start of this problem (Example 8.9), $A(z)$ was **defined** as:

$$A(z) = a_0 + a_1 z + a_2 z^2 + \cdots + a_n z^n + \cdots = \sum_{i \geq 0} a_i z^i$$

That is: the coefficient sitting in front of z^i in $A(z)$ is, **by definition**, a_i.

What We Just Derived

The image shows that after all the algebra (partial fractions, expanding each piece as a geometric series, and adding them together), we arrived at:

$$A(z) = \sum_{i \geq 0} (2^{i+1} - 1) z^i$$

Match the Two Expressions Directly

We have two expressions for the *same* $A(z)$:

$$\sum_{i \geq 0} a_i z^i \quad = \quad \sum_{i \geq 0} (2^{i+1} - 1) z^i$$

Since these are the same series, the coefficients of z^i on each side **must be equal**, term by term:

$$a_i = 2^{i+1} - 1$$

Relabel $i \to n$

The index letter (i vs. n) is just a dummy variable — it doesn't matter which letter we use, so we can freely rename it back to n to match the original problem's notation:

$$a_n = 2^{n+1} - 1$$

Why This "Read Off the Coefficient" Step Is Always the Last Move

This is the payoff of the entire generating-function method, worth stating explicitly to students: **the whole point of building $A(z)$ and manipulating it algebraically is to eventually rewrite it in the form $\sum a_i z^i$ with an explicit, recognizable formula in place of a_i.** Once you've done that, you're done — there's no further work, because the coefficient of z^i *is*, by the original definition, the sequence term a_i you were solving for.

Cross-Check Against Earlier Work

This matches exactly what you found via substitution earlier in this conversation for the same recurrence ($a_n = 2a_{n-1} + 1$, $a_0 = 1$):

$$a_n = 2^{n+1} - 1$$

Same formula, confirming both methods agree. Good example to show students side-by-side as a "two roads, same destination" illustration.

Generating Functions

Generating functions are used to represent sequences by coding the terms of a sequence as coefficients of powers of a variable z in a formal power series. $A(z) = a_0 + a_1 z + a_2 z^2 + ... + a_n z^n + ... = \sum_{i=0} a_i z^i$

- The above equation is referred to as the ***ordinary generating function*** of $\{a_n\}$

Generating functions can be used to solve a recurrence relations by translating a recurrence relation for the terms of a sequence into an equation involving a generating function. This equation can then be solved to find the closed form for the generating function.

Study Materials: Generating Functions

Three sections as requested, each with explanation, worked examples, and practice problems (with answer keys) at the end.

Part 1: Generating Functions

1.1 Definition

The **ordinary generating function (OGF)** of a sequence $a_0, a_1, a_2, \ldots$ is the formal power series:

$$A(z) = \sum_{n=0}^{\infty} a_n z^n = a_0 + a_1 z + a_2 z^2 + \cdots$$

Key conceptual point: z is a formal placeholder, not a number being plugged in. We never ask "does this converge?" — we only care about which coefficient sits with which power of z. Think of $A(z)$ as a clothesline: each a_n is hung at position z^n.

1.2 Building Generating Functions from Scratch

Example A: Sequence $a_n = 1$ for all n (i.e., $1, 1, 1, 1, \ldots$)

$$A(z) = 1 + z + z^2 + z^3 + \cdots = \frac{1}{1 - z}$$

This is the single most important building block — nearly everything else derives from it.

Example B: Sequence $a_n = r^n$ (i.e., $1, r, r^2, r^3, \ldots$)

$$A(z) = 1 + rz + r^2z^2 + \cdots = \sum_{n \geq 0} (rz)^n = \frac{1}{1 - rz}$$

Example C: Sequence $a_n = c$ (a fixed constant, for all n)

$$A(z) = c + cz + cz^2 + \cdots = c \cdot \frac{1}{1 - z} = \frac{c}{1 - z}$$

Example D: Finite sequence $a_0 = 1, a_1 = 2, a_2 = 3$, and $a_n = 0$ for $n \geq 3$

$$A(z) = 1 + 2z + 3z^2$$

Worth noting: finite sequences just give **polynomials**, not infinite series — a good simple case to anchor the concept before moving to infinite ones.

1.3 Standard Generating Functions Table (memorize/derive these)

Sequence a_n	Generating Function
$1, 1, 1, \ldots$	$\frac{1}{1 - z}$
$1, r, r^2, \ldots$	$\frac{1}{1 - rz}$
$1, 0, 1, 0, \ldots$	$\frac{1}{1 - z^2}$
$0, 1, 2, 3, \ldots$ (i.e. $a_n = n$)	$\frac{z}{(1 - z)^2}$
$\binom{n+k-1}{k-1}$	$\frac{1}{(1 - z)^k}$

Practice Problems — Part 1

P1.1 Write the generating function for the sequence $a_n = 5$ for all $n \geq 0$.

P1.2 Write the generating function for the sequence $1, -1, 1, -1, \ldots$ (i.e., $a_n = (-1)^n$).

P1.3 Write the generating function for the sequence $2, 6, 18, 54, \ldots$ (i.e., $a_n = 2 \cdot 3^n$).

P1.4 Write out the generating function (as a polynomial) for the finite sequence $a_0 = 4, a_1 = 0, a_2 = 7$, and $a_n = 0$ for $n \geq 3$.

Answer Key — Part 1

P1.1: $A(z) = \dfrac{5}{1-z}$

P1.2: This is $r = -1$ in Example B: $A(z) = \dfrac{1}{1-(-1)z} = \dfrac{1}{1+z}$

P1.3: This is $2 \cdot r^n$ with $r = 3$: $A(z) = \dfrac{2}{1-3z}$

P1.4: $A(z) = 4 + 7z^2$

Part 2: Function of Sequences (Operations on Generating Functions)

This section covers how operations on a sequence translate into operations on its generating function. Given $A(z)$ for $\{a_n\}$ and $B(z)$ for $\{b_n\}$:

2.1 Addition and Scalar Multiplication

If $c_n = a_n + b_n$: $\quad C(z) = A(z) + B(z)$

If $c_n = k \cdot a_n$: $\quad C(z) = k \cdot A(z)$

Example: Find the GF for $c_n = 3 + 2^n$.

3 has GF $\dfrac{3}{1-z}$; 2^n has GF $\dfrac{1}{1-2z}$. So:

$$C(z) = \frac{3}{1-z} + \frac{1}{1-2z}$$

2.2 Right-Shift (Prepend Zeros)

If $b_n = a_{n-k}$ for $n \geq k$, and $b_n = 0$ for $n < k$:

$$B(z) = z^k A(z)$$

Why: shifting the sequence right by k places means every term's power of z increases by k — exactly what multiplying by z^k does.

Example: If $a_n = 1, 1, 1, \ldots$ (GF $= \frac{1}{1-z}$), then $b_n = 0, 0, 1, 1, 1, \ldots$ (two leading zeros) has GF:

$$B(z) = z^2 \cdot \frac{1}{1-z} = \frac{z^2}{1-z}$$

2.3 Left-Shift (Drop Leading Terms)

If $b_n = a_{n+k}$ (i.e., we chop off the first k terms and shift everything left):

$$B(z) = \frac{A(z) - a_0 - a_1 z - \cdots - a_{k-1} z^{k-1}}{z^k}$$

Example: If $a_n = 1, 2, 3, 4, 5, \ldots$ starting $a_0 = 1$, and $b_n = a_{n+1}$ (drop the first term), then:

$$B(z) = \frac{A(z) - a_0}{z}$$

2.4 Multiplication by n

If $b_n = n \cdot a_n$:

$$B(z) = z, A'(z)$$

Why this works: differentiating $A(z) = \sum a_n z^n$ term-by-term gives $A'(z)$ $= \sum n, a_n z^{n-1}$; multiplying by z restores the correct power, giving $\sum n, a_n z^n$.

Example: Since $\frac{1}{1-z}$ generates $1, 1, 1, \ldots$, differentiate: $\left(\frac{1}{1-z}\right)' = \frac{1}{(1-z)^2}$. Multiply by z:

$$z \cdot \frac{1}{(1-z)^2} = \frac{z}{(1-z)^2}$$

which correctly generates $a_n = n$ (i.e., $0, 1, 2, 3, \ldots$) — matching the table entry from Part 1.

2.5 Convolution (the most important operation)

If $c_n = \sum_{k=0}^{n} a_k b_{n-k}$ (the **convolution** of the two sequences):

$$C(z) = A(z) \cdot B(z)$$

Why this matters so much: multiplying two generating functions automatically computes this sum for you — this is the single biggest reason GFs are powerful for counting problems.

Example: Let $a_n = 1$ for all n (GF $\frac{1}{1-z}$) and $b_n = 1$ for all n (same GF). Then:

$$C(z) = \frac{1}{1-z} \cdot \frac{1}{1-z} = \frac{1}{(1-z)^2}$$

By the table, $\frac{1}{(1-z)^2}$ generates $c_n = n + 1$. Check directly: $c_n = \sum_{k=0}^{n} 1 \cdot 1 = n + 1$ ✓ (there are $n+1$ ways to pick k from 0 to n).

2.6 Summation (Partial Sums)

If $b_n = \sum_{k=0}^{n} a_k$ (running total of a):

$$B(z) = \frac{A(z)}{1-z}$$

Why: this is really convolution with the all-ones sequence, since $\frac{1}{1-z}$ is the GF for $1, 1, 1, \ldots$, and convolving with all-ones just sums up all prior terms.

Example: $a_n = 1$ for all n has GF $\frac{1}{1-z}$. The partial sums are $b_n = n+1$. Using the rule:

$$B(z) = \frac{1/(1-z)}{1-z} = \frac{1}{(1-z)^2}$$

— matches the convolution example above (same result, different route).

Practice Problems — Part 2

P2.1 Given $a_n = 1, 1, 1, \ldots$ with GF $\dfrac{1}{1-z}$, find the GF for $b_n = a_{n-3}$ (shift right by 3, with $b_0 = b_1 = b_2 = 0$).

P2.2 Find the GF for $c_n = 4 \cdot a_n + 2 \cdot b_n$, where $A(z) = \dfrac{1}{1-z}$ and $B(z) = \dfrac{1}{1-2z}$.

P2.3 Using the "multiply by n" rule, find the GF for $a_n = n \cdot 2^n$. (Start from the GF for 2^n.)

P2.4 Find the GF for $c_n = \sum_{k=0}^{n} 2^k$ (partial sums of 2^n), using the summation rule. Then simplify and check against the direct formula $c_n = 2^{n+1} - 1$ from earlier in our conversation.

P2.5 If $a_0 = 1, a_1 = 2, a_2 = 3, a_3 = 4, \ldots$ (i.e., $a_n = n+1$), find the GF for $b_n = a_{n+1}$ (left-shift by 1).

Answer Key — Part 2

P2.1: $B(z) = z^3 \cdot \dfrac{1}{1-z} = \dfrac{z^3}{1-z}$

P2.2: $C(z) = \dfrac{4}{1-z} + \dfrac{2}{1-2z}$

P2.3: $A(z)$ for 2^n is $\dfrac{1}{1-2z}$. Differentiate: $\dfrac{d}{dz}\left(\dfrac{1}{1-2z}\right) = \dfrac{2}{(1-2z)^2}$. Multiply by z:

$$B(z) = \frac{2z}{(1-2z)^2}$$

P2.4: $A(z) = \dfrac{1}{1-2z}$ (GF for 2^n). Apply summation rule:

$$C(z) = \frac{1/(1-2z)}{1-z} = \frac{1}{(1-2z)(1-z)}$$

This is the exact same $A(z)$ from Example 8.9 (after partial fractions, we already showed it expands to $2^{n+1}-1$) — good cross-check that the summation rule reproduces a result you've already verified.

P2.5: $A(z) = \dfrac{1}{(1-z)^2}$ (GF for $n+1$, from the table). Left-shift by 1:

$$B(z) = \frac{A(z) - a_0}{z} = \frac{\frac{1}{(1-z)^2} - 1}{z}$$

Part 3: Calculating Coefficients of Generating Functions

Once you have a closed-form $A(z)$, the goal is extracting $a_n = [z^n]A(z)$ (read: "the coefficient of z^n in $A(z)$").

3.1 Method 1 — Generalized Binomial Series

$$\frac{1}{(1-z)^k} = \sum_{n=0}^{\infty} \binom{n+k-1}{k-1} z^n$$

So if $A(z) = \dfrac{1}{(1-z)^k}$, then $a_n = \binom{n+k-1}{k-1}$.

Example: $A(z) = \dfrac{1}{(1-z)^3}$. Here $k=3$, so $a_n = \binom{n+2}{2} = \dfrac{(n+2)(n+1)}{2}$.

Check $n=0$: $a_0 = \binom{2}{0}$\$...*careful*—\$$\binom{2}{2} = 1$, matches $A(z)$'s constant term of 1 ✓.

3.2 Method 2 — Partial Fraction Decomposition (the main workhorse)

This is essential whenever $A(z)$ is a rational function $\dfrac{P(z)}{Q(z)}$ — exactly what recurrence relations produce.

Procedure:

1. Factor $Q(z)$ into linear factors (and irreducible quadratics, if any)
2. Decompose: $\dfrac{P(z)}{Q(z)} = \dfrac{A}{1-r_1 z} + \dfrac{B}{1-r_2 z} + \cdots$
3. Expand each piece as a geometric series: $\dfrac{1}{1-rz} = \sum r^n z^n$

4. Add up contributions to get a_n

Worked Example (full, from scratch): $A(z) = \dfrac{1}{(1-z)(1-4z)}$

Set $\dfrac{1}{(1-z)(1-4z)} = \dfrac{P}{1-z} + \dfrac{Q}{1-4z}$

Clear denominators: $1 = P(1-4z) + Q(1-z)$

At $z=1$: $1 = P(1-4) \implies P = -\frac{1}{3}$

At $z=\frac{1}{4}$: $1 = Q(1-\frac{1}{4}) \implies 1 = Q\cdot\frac{3}{4} \implies Q = \frac{4}{3}$

$$A(z) = \frac{-1/3}{1-z} + \frac{4/3}{1-4z} = \sum_{n\geq 0}\left(-\frac{1}{3} + \frac{4}{3}\cdot 4^n\right) z^n$$

$$a_n = \frac{4}{3}\cdot 4^n - \frac{1}{3} = \frac{4^{n+1}-1}{3}$$

Check: a_0 should be 1 (constant term of $A(z)$). Formula: $\frac{4-1}{3} = 1$ ✓

3.3 Method 3 — Repeated Roots

If $(1-rz)^k$ appears as a factor (a repeated root):

$$\frac{1}{(1-rz)^k} = \sum_{n\geq 0}\binom{n+k-1}{k-1} r^n z^n$$

Example: $A(z) = \dfrac{1}{(1-2z)^2}$. Here $r=2, k=2$:

$$a_n = \binom{n+1}{1}\cdot 2^n = (n+1)2^n$$

This is worth connecting back to your earlier repeated-root characteristic-equation example ($a_n = 4a_{n-1} - 4a_{n-2}$, giving $a_n = (n+1)2^n$) — the generating function for that recurrence works out to exactly $\dfrac{1}{(1-2z)^2}$ (times a constant depending on initial conditions), and this repeated-root coefficient formula is the GF-side counterpart of the "multiply by n" fix you used there.

Practice Problems — Part 3

P3.1 Find a_n if $A(z) = \dfrac{1}{(1-z)^4}$.

P3.2 Find a_n if $A(z) = \dfrac{1}{(1-3z)(1-z)}$ (use partial fractions).

P3.3 Find a_n if $A(z) = \dfrac{2}{(1-z)(1-2z)}$ (note the numerator is 2, not 1 — adjust your partial fraction setup accordingly).

P3.4 Find a_n if $A(z) = \dfrac{1}{(1-3z)^2}$ (repeated root case).

P3.5 (challenge) Find a_n if $A(z) = \dfrac{z}{(1-z)(1-2z)}$ (note the extra z in the numerator — after partial-fractioning the denominator part, remember to account for the leftover factor of z, which right-shifts the sequence by one).

Answer Key — Part 3

P3.1: $k=4$: $a_n = \binom{n+3}{3} = \dfrac{(n+3)(n+2)(n+1)}{6}$

P3.2: Set $\dfrac{1}{(1-3z)(1-z)} = \dfrac{P}{1-3z} + \dfrac{Q}{1-z}$. At $z=\frac{1}{3}$: $1 = P(1-\frac{1}{3}) \implies P = \frac{3}{2}$. At $z=1$: $1 = Q(1-3) \implies Q = -\frac{1}{2}$.

$$a_n = \frac{3}{2}\cdot 3^n - \frac{1}{2} = \frac{3^{n+1}-1}{2}$$

P3.3: Set $\dfrac{2}{(1-z)(1-2z)} = \dfrac{P}{1-z} + \dfrac{Q}{1-2z}$. At $z=1$: $2 = P(1-2) \implies P = -2$. At $z=\frac{1}{2}$: $2 = Q(1-\frac{1}{2}) \implies Q = 4$.

$$a_n = 4\cdot 2^n - 2 = 2^{n+2} - 2$$

P3.4: $r=3, k=2$: $a_n = \binom{n+1}{1}\cdot 3^n = (n+1)3^n$

P3.5: First solve for the coefficients of $\dfrac{1}{(1-z)(1-2z)}$ (from P2.4's setup — same as Example 8.9's structure): $[z^n] = 2^{n+1} - 1$. Since the actual function has an extra factor of z, this right-shifts the whole sequence by one place (recall Section 2.2: multiplying by z^1 shifts right by 1). So:

$$a_n = \begin{cases} 0 & n = 0 \\ 2^n - 1 & n \geq 1 \end{cases}$$

(Check by relabeling: the coefficient of z^n in $z\cdot A(z)$ equals the coefficient of z^{n-1} in $A(z)$, which is $2^{(n-1)+1} - 1 = 2^n - 1$, for $n \geq 1$.)